

QUESTIONS 31 / 31 completed

Arithmetic

ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class main{
3     public static void main(String[] args)
4     {
5         int a,b;
6         char ch;
7         Scanner s1=new Scanner(System.in);
8         System.out.println("Enter 2 values");
9         a=s1.nextInt();
10        b=s1.nextInt();
11        System.out.println("A->Addition,S->Subtraction,M->Multiplication");
12        ch=s1.next().charAt(0);
13        switch(ch)
14        {
15            case'A':System.out.println(a+b);
16            break;
17            case'B':System.out.println(a-b);
18            break;
19            case'C':System.out.println(a*b);
20            break;
21        }
22    }
23 }
```

OUTPUT

```
Enter 2 values
A->Addition,S->Subtraction,M->Multiplication
14
```

INPUT

5
9
A

QUESTIONS 31 / 31 completed

MAX

MAX

Write a Java program to find the maximum of three numbers using conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Maxnum{
3     public static void main(String[] args){
4         int a=25;
5         int b=30;
6         int c=60;
7
8         if(a>b){
9             if(a>c)
10                System.out.println("a is the greatest");
11            else
12                System.out.println("b is the greatest");
13        }
14        else{
15            if(b>c)
16                System.out.println("b is the greatest");
17            else
18                System.out.println("c is the greatest");
19        }
20    }
21 }
```

OUTPUT

```
c is the greatest
```

QUESTIONS 31 / 31 completed

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 class main{
3     public static void main(String[] args){
4         Scanner S=new Scanner(System.in);
5         int a=S.nextInt();
6         if((a%4==0 && a%100 !=0)|| (a%400==0)){
7             System.out.println("It is a Leap Year");
8         }
9         else{
10            System.out.println("It is not a leap year");
11        }
12    }
13 }
```

OUTPUT

```
It is a Leap Year
```

INPUT

2024

QUESTIONS 31 / 31 completed

Fibonacci

FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 class main{
3 public static void main(String[] args){
4     Scanner S=new Scanner(System.in);
5     int n=S.nextInt();
6     int a=0;
7     int b=1;
8     System.out.print(a+" "+b+" ");
9     for (int i=2;i<n;i++){
10         int c=a+b;
11         System.out.print(c+" ");
12         a=b;
13         b=c;
14     }
15 }
16 }
```

OUTPUT

0112358

INPUT

5

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class main{
3 public static void main(String[] args){
4     Scanner S=new Scanner(System.in);
5     int n=S.nextInt();
6
7     int fact=1;
8     for(int i=1;i<=n;i++){
9         fact = fact*i;
10    }
11    System.out.println("Factorial of " + n + " is : " + fact);
12 }
13 }
```

OUTPUT

Factorial of 5 is : 120

INPUT

5

QUESTIONS 31 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:

```
***
***
****
*****
```

PROGRAM EDITOR

Java

Run

Save

```
1 class main{
2 public static void main(String[] args){
3     int n=5;
4     for (int i=1;i<=n;i++) {
5         for (int j=1;j<=i;j++){
6             System.out.print("**");
7         }
8         System.out.println();
9     }
10 }
11 }
```

OUTPUT

```
***
***
****
*****
```

QUESTIONS 31 / 31 completed

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner S1=new Scanner(System.in);
5
6         int n=S1.nextInt();
7         int[] arr=new int[n];
8         for (int i=0;i<n;i++)arr[i]=S1.nextInt();
9
10        int key=S1.nextInt();
11        int pos=Arrays.binarySearch(arr,key);
12        if(pos<0)
13            System.out.println("Element not found");
14        else
15            System.out.println("Element Found at index : "+pos);
16    }
17 }
```

OUTPUT

Element Found at index : 3

INPUT

5
10 20 30 40 50
40

QUESTIONS 31 / 31 completed

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner S1=new Scanner(System.in);
5         int a = S1.nextInt();
6         int b = S1.nextInt();
7
8         while(b!= 0){
9             int temp=b;
10            b=a%b;
11            a=temp;
12        }
13        System.out.println("GCD : "+a);
14    }
15 }
```

OUTPUT

GCD : 12

INPUT

36
60

QUESTIONS 31 / 31 completed

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int n =s.nextInt();
6         boolean prime=true;
7
8         if(n<=1)prime=false;
9         else{
10             for(int i=2;i<=n/2;i++){
11                 if(n%i==0){
12                     prime=false;
13                     break;
14                 }
15             }
16         }
17
18         if(prime)
19             System.out.println("prime");
20         else
21             System.out.println("not prime");
22     }
```

OUTPUT

prime

INPUT

7

QUESTIONS 31 / 31 completed

Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int n=s.nextInt();
6         int[] arr=new int[n];
7         for(int i=0;i<n;i++){
8             arr[i]=s.nextInt();
9         }
10        for(int i=0;i<n-1;i++){
11            int min=i;
12            for(int j=i+1;j<n;j++){
13                if(arr[j]<arr[min])min=j;
14            }
15            int temp=arr[min];
16            arr[min]=arr[i];
17            arr[i]=temp;
18        }
19        for(int i=0;i<n;i++){
20            System.out.print(arr[i]+" ");
21        }
22    }
23 }
```

OUTPUT

3 21 32 45 64

INPUT

5
32
45
21
64
3

QUESTIONS 31 / 31 completed

Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int n=s.nextInt();
6         for(int i=1;i<=10;i++){
7             System.out.println(n+" X "+i+" = "+ (n*i));
8         }
9     }
10 }
```

OUTPUT

5 X 1 = 5
5 X 2 = 10
5 X 3 = 15
5 X 4 = 20
5 X 5 = 25
5 X 6 = 30

INPUT

5

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     static int fact(int n){
4         if(n==0||n==1)return 1;
5         return n*fact(n-1);
6     }
7     public static void main(String[] args){
8         Scanner s=new Scanner(System.in);
9         int n=s.nextInt();
10        System.out.println("Factorial : "+fact(n));
11    }
12 }
```

OUTPUT

Factorial : 120

INPUT

5

QUESTIONS 31 / 31 completed

LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops:*****

```
*****
****
***
**
*
```

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main{
2     public static void main(String[] args){
3         for(int i=5;i>=1;i--){
4             for(int j=1;j<=i;j++){
5                 System.out.print("**");
6             }
7             System.out.println();
8         }
9     }
10 }
```

OUTPUT

```
*****
****
***
**
*
```

QUESTIONS 31 / 31 completed

LCM

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int a=s.nextInt();
6         int b=s.nextInt();
7         int lcm=(a>b) ? a:b;
8
9         while(true){
10             if(lcm%a==0 && lcm%b==0){
11                 System.out.println("LCM : "+lcm);
12                 break;
13             }
14             lcm++;
15         }
16     }
17 }
```

OUTPUT

```
LCM : 36
```

INPUT

```
12
18
```

QUESTIONS 31 / 31 completed

D to B

D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int[] bin = new int[32];
8         int i = 0;
9
10        while (n > 0) {
11            bin[i++] = n % 2;
12            n = n / 2;
13        }
14
15        for (int j = i - 1; j >= 0; j--) {
16            System.out.print(bin[j]);
17        }
18    }
19 }
20
```

OUTPUT

```
1010
```

INPUT

```
10
```

QUESTIONS 31 / 31 completed

B to D

B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         String bin = sc.next();
7         int dec = 0, pow = 1;
8
9         for (int i = bin.length() - 1; i >= 0; i--) {
10             if (bin.charAt(i) == '1') dec += pow;
11             pow *= 2;
12         }
13
14         System.out.println("Decimal: " + dec);
15     }
16 }
```

OUTPUT

Decimal: 10

INPUT

1010

QUESTIONS 31 / 31 completed

Sum

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[]args){
4         Scanner s=new Scanner(System.in);
5         int n=s.nextInt();
6         int sum=0;
7         while(n>0){
8             sum +=n%10;
9             n=n/10;
10        }
11        System.out.println(sum);
12    }
13 }
```

OUTPUT

10

INPUT

1234

QUESTIONS 31 / 31 completed

Product

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[]args){
4         Scanner s=new Scanner(System.in);
5         int n=s.nextInt(),prod=1;
6         while(n>0){
7             prod*=n%10;
8             n/=10;
9         }
10        System.out.println(prod);
11    }
12 }
13 }
```

OUTPUT

24

INPUT

1234

QUESTIONS 31 / 31 completed

Power

POWER

Write a Java program to find the power of a number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int base=s.nextInt();
6         int exp=s.nextInt();
7         int result=1;
8
9         for(int i=1;i<=exp;i++){
10             result *=base;
11         }
12         System.out.println(result);
13     }
14 }
```

OUTPUT

32

INPUT

2
5

QUESTIONS 31 / 31 completed

Area Circle

AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         double r=s.nextDouble();
6         double area=Math.PI *r*r;
7         System.out.println(area);
8     }
9 }
```

OUTPUT

78.53981633974483

INPUT

5

QUESTIONS 31 / 31 completed

Shpere

SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         double r=s.nextDouble();
6         double volume=(4.0/3)*Math.PI*r*r*r;
7         System.out.println(volume);
8     }
9 }
```

OUTPUT

113.09733552923255

INPUT

3

QUESTIONS 31 / 31 completed

Palindrome

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         String str=s.nextLine();
6         String rev="";
7         for(int i=str.length()-1;i>=0;i--){
8             rev=rev+str.charAt(i);
9         }
10        if(str.equals(rev)){
11            System.out.println("its a palindrome");
12        }
13        else{
14
15
16            System.out.println("Its is not a palindrome");
17        }
18    }
19 }
```

OUTPUT

its a palindrome

INPUT

malayalam

QUESTIONS 31 / 31 completed

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int temp = n, sum = 0;
7         int digits = String.valueOf(n).length();
8         while (temp > 0) {
9             int d = temp % 10;
10            int pow = 1;
11            for (int i = 0; i < digits; i++) {
12                pow *= d;
13            }
14            sum += pow;
15            temp /= 10;
16        }
17
18        if (sum == n) {
19            System.out.println("Armstrong");
20        } else {
21            System.out.println("Not Armstrong");
22        }
23    }
24 }
```

OUTPUT

INPUT

153

QUESTIONS 31 / 31 completed

RT-num.Pattern

RT- NUM.PATTERN

Write a Java program to print the following pattern using loops:

```
1
22
333
4444
55555
```

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         for (int i = 1; i <= n; i++) {
7             for (int j = 1; j <= i; j++) {
8                 System.out.print(i);
9             }
10            System.out.println();
11        }
12    }
13 }
14 }
```

OUTPUT

```
1
22
333
4444
55555
```

INPUT

5

QUESTIONS 31 / 31 completed

Perfect

PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int sum = 0;
7         for (int i = 1; i < n; i++) {
8             if (n % i == 0) {
9                 sum += i;
10            }
11        }
12
13        if (sum == n) {
14            System.out.println("Perfect Number");
15        } else {
16            System.out.println("Not Perfect Number");
17        }
18    }
19 }
20
```

OUTPUT

Perfect Number

INPUT

28

QUESTIONS 31 / 31 completed

Max Array

MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] arr = new int[n];
7         for (int i = 0; i < n; i++) arr[i] = sc.nextInt();
8         int max = arr[0];
9         for (int i = 1; i < n; i++) {
10             if (arr[i] > max) max = arr[i];
11         }
12         System.out.println("Largest element: " + max);
13     }
14 }
15
```

OUTPUT

Largest element: 64

INPUT

4
12
43
35
64

QUESTIONS 31 / 31 completed

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         for (int i = 0; i < n; i++) arr[i] = sc.nextInt();
9         for (int i = 0; i < n - 1; i++) {
10             for (int j = 0; j < n - i - 1; j++) {
11                 if (arr[j] > arr[j + 1]) {
12                     int temp = arr[j];
13                     arr[j] = arr[j + 1];
14                     arr[j + 1] = temp;
15                 }
16             }
17         }
18
19         System.out.println("Second largest element: " + arr[n - 2]);
20     }
21 }
```

OUTPUT

Second largest element: 21

INPUT

5
10
21
32
21
20

QUESTIONS 31 / 31 completed

Min

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         int[] arr = {10, 25, 7, 30, 18};
4         int min = arr[0];
5         for (int i = 1; i < arr.length; i++) {
6             if (arr[i] < min) {
7                 min = arr[i];
8             }
9         }
10        System.out.println("Smallest element: " + min);
11    }
12 }
13
14
```

OUTPUT

Smallest element: 7

QUESTIONS 31 / 31 completed

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         int[] arr = {10, 25, 7, 30, 18};
4         int min = arr[0];
5         for (int i = 1; i < arr.length; i++) {
6             if (arr[i] < min) min = arr[i];
7         }
8         int second = Integer.MAX_VALUE;
9         for (int i = 0; i < arr.length; i++) {
10            if (arr[i] != min && arr[i] < second) {
11                second = arr[i];
12            }
13        }
14        if (second == Integer.MAX_VALUE) {
15            System.out.println("No second smallest element");
16        } else {
17            System.out.println("Second smallest element: " + second);
18        }
19    }
20 }
21
```

OUTPUT

Second smallest element: 10

QUESTIONS 31 / 31 completed

Sort-B

SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         int[] arr = {10, 25, 7, 30, 18};
4         int n = arr.length;
5         for (int i = 0; i < n - 1; i++) {
6             for (int j = 0; j < n - i - 1; j++) {
7                 if (arr[j] > arr[j + 1]) {
8                     int temp = arr[j];
9                     arr[j] = arr[j + 1];
10                    arr[j + 1] = temp;
11                }
12            }
13        }
14        System.out.print("Sorted array: ");
15        for (int i = 0; i < n; i++) {
16            System.out.print(arr[i] + " ");
17        }
18    }
19 }
20
```

OUTPUT

Sorted array: 7 10 18 25 30

QUESTIONS 31 / 31 completed

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         int[] arr = {10, 25, 7, 30, 18};
4         int n = arr.length;
5         for (int i = 0; i < n - 1; i++) {
6             int min = i;
7             for (int j = i + 1; j < n; j++) {
8                 if (arr[j] < arr[min]) min = j;
9             }
10            int temp = arr[min];
11            arr[min] = arr[i];
12            arr[i] = temp;
13        }
14        for (int x : arr) System.out.print(x + " ");
15    }
16 }
17
18
19
```

OUTPUT

Sorted array: 7 10 18 25 30